

CONSENT FORM FOR ORIGINALITY OF IDEA

School Name: School of Computer Science and Engineering

Group Number (IN Excel file mentioned): 27

Date:

Title of Idea/Project: AI-Powered Smart Helmet for Accident Prevention and Emergency Response

I/We, hereby affirm that the idea titled above is our original idea and has not been copied, borrowed, or directly reproduced from any other source without proper acknowledgment.

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5. I/We do not wish to share the invention details with any third party other than those directly involved in the patent filing process, ensuring strict confidentiality.

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Pre-Assessment Patent Idea Submission

Patent Idea Title

**AI-Powered Smart Helmet for Accident Prevention and
Emergency Response**

Submitted By

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In partial fulfilment for the requirements of the award for the degree of
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SUBMITTED TO

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Problem Statement & Justification

Road accidents, especially those involving two-wheelers, are a leading cause of death and serious injuries worldwide. One of the biggest issues is the delay in getting medical help after an accident, which greatly reduces the chances of survival. Many victims suffer because they do not receive emergency aid on time.

Another major problem is that many riders avoid wearing helmets, either because they find them uncomfortable or simply ignore the risk. Even those who wear helmets do not have access to smart safety features that could help prevent accidents or send alerts in case of an emergency.

Reckless riding behaviours, such as speeding, sudden braking, and breaking traffic rules, also contribute to accidents. These dangerous habits often go unnoticed until it's too late. Additionally, hit-and-run cases are increasing, leaving victims stranded without a way to call for help.

Problem significance

- **High Death Rates:** Many lives could be saved if medical help reached accident victims faster.
- **No Real-Time Monitoring:** Regular helmets don't track a rider's condition or location during an accident.
- **Low Helmet Usage:** Many people don't wear helmets, increasing the risk of fatal injuries. A smart helmet with safety features could encourage more people to wear them.
- **Delayed Emergency Response:** Even if an accident is reported, finding and reaching the victim takes time. A system that shares the exact location can speed up response times.
- **Workplace Safety Risks:** In jobs like construction and mining, helmets are essential for protection, but there's no smart system to monitor worker safety in real-time.

An AI-powered smart helmet enhances safety, promotes helmet use, and enables rapid emergency response, reducing fatalities.

Novelty & Uniqueness

While smart helmets are already offered in the market, most of the products available are crash detection and Bluetooth connectivity-based. They lack a stable verification system and tend to provide false alarms in case of sudden movement or harsh braking.

Our AI-powered smart helmet stands out by integrating multiple advanced technologies, including:

- **Accident Verification System** cross-verifying real-time data from different sensors such as rider movement, helmet impact force, and GPS tracking to prevent false alarms and send timely emergency notifications.
- **AI-based accident prevention** that continuously tracks the behavior of riders and road conditions to identify likely dangers before the accident takes place.
- **IoT-based emergency response** that automatically sends alerts with the precise GPS coordinates to family and friends and emergency medical personnel in case of a confirmed crash.
- **Real-time impact analysis** that cleverly distinguishes between minor shocks and harsh crashes in order not to send false alarms.
- **Voice-controlled operation** for hands-free conversation, emergency messages, and safety notices, providing convenience and driver concentration.

By integrating these capabilities, our intelligent helmet offers an end-to-end safety solution above simple crash detection, improving both accident prevention and emergency response.

Technical Details & Implementation

Imagine wearing a helmet that doesn't just sit on your head for protection but actually thinks and reacts in real time to keep you safe. Our AI-Powered Smart Helmet does exactly that! It's like having a personal safety assistant that monitors your ride, detects danger, and alerts help when needed.

How it works?

This helmet is packed with tiny smart sensors (like those in your smartphone) that constantly track your movement. If you suddenly fall, crash, or make an unusual movement, the sensors immediately detect it. Then, the AI (Artificial Intelligence) brain inside the helmet analyses the situation and decides if an accident has actually happened. If it has, the helmet automatically sends an emergency alert to your family, friends, or even emergency services with your exact location using GPS tracking.

Now, imagine you're riding late at night and suddenly feel dizzy. The helmet can detect unusual motion patterns and even warn you to take a break if it senses risky behaviour like over-speeding or drowsiness.

Key Components

Smart Brain

- The helmet has a tiny smart brain that watches how you ride.
- It knows the difference between a small bump and a real accident.

Movement Sensors

- These sensors feel every sudden movement, like a hard fall or sharp turn.
- If something unusual happens, the helmet checks if it's an accident.

GPS & Mobile Connection

- Always knows where you are and shares your location in case of an accident.
- If you crash, it automatically sends an SOS to emergency contact

Emergency Alert System

- Stores your ride details and sends alerts even if you can't.
- Works instantly, even if you're unconscious.

Voice Commands & Smart Reminders

- You can talk to your helmet to call someone or ask for help.
- It can remind you to wear it before starting the ride.

Long-Lasting Battery

- Designed to last long, so you don't have to charge it often

Specifications

Stops accidents before they happen by warning you about risky riding.

Sends help automatically when you can't do it yourself.

Encourages safety by reminding riders to wear helmets and ride carefully.

Potential Applications & Market Scope

Who can wear this smart helmet?

- Bike and recent cyclists: The feature would generate an automatic call whenever there is an accident, keeping riders safe.
- Traffic police and officers preach road safety and responsibility in riding.
- These alert hospitals or ambulances immediately in cases of accidents, whereby emergency services contact hospitals and ambulances.
- Workers of hazardous occupations: Such as mining, construction, factories can detect falls and injuries and notify the managers for effective and immediate action.

Where can I use this Smart Helmet?

- Road safety and accident avoidance: This makes monitoring dangerous riding behaviors, alerting emergencies, hence reducing the number of accidents.
- This safety monitoring at workplace identifies falls to realize injuries that are most likely to happen among employees in a risky environment such as mines and construction sites.
- Insurance and healthcare: It bears hospitals a fast emergency response and helps insurance firms with the risk the victims had before the accident happened.

Feasibility & Implementation Plan

Resources Needed for Implementation

Hardware:

- Computers, servers, and networking equipment tailored to project needs.
- Specialized devices or machinery, if applicable.

Software:

- Project management tools (e.g., Wrike, Jira, or MS Project).
- Industry-specific software required for execution (e.g., CAD tools for design projects).

Data Sources:

- Access to relevant databases or APIs for information retrieval.
- Market research data and analytics tools.

Funding:

- Budget allocation for hardware, software, personnel, and unforeseen costs.
- Potential external funding sources like grants or investors.

Challenges & Limitations**Potential Risks:**

- Technical challenges due to outdated or incompatible technology.
- Financial risks like budget overruns or insufficient funding.

Constraints:

- Legal and regulatory compliance issues (e.g., permits, data protection laws).
- Resource availability, including skilled personnel and equipment shortages.

Difficulties in Implementation:

- Resistance from stakeholders or lack of buy-in.
- Coordination challenges among cross-functional teams.

Timeline for Development

Research Phase: 2-4 months to gather data, assess feasibility, and finalize requirements.

Prototyping Phase: 3-6 months to develop initial models and conduct iterative testing.

Testing Phase: 2-3 months for quality assurance and user feedback loops.

Implementation Phase: 6-12 months for full-scale deployment and monitoring milestones

Ethical & Legal Considerations

Confidentiality and data protection

Privacy concerns arise in this case because our smart helmet employs artificial intelligence (AI) to collect personal data such as the location, movement history, and health information of an accident-stricken rider. Encryption methods will be employed to safeguard user data in a way that all the information being stored and transmitted is secure and not readable by unauthorized third parties.

Besides, before we collect any user personal data, we will get their explicit consent. With permission control, clearing of the cached data, or exclusion from specific functions, the users will also be in complete control of their data. In regards to safeguarding personal data, our system will also be in line with international data protection legislation such as the India Digital Personal Data Safety Act (DPDP Act) and the General Data Protection Regulation (GDPR). To ensure that our helmets comply with every requirement of the law, continual product testing and periodic checks of road safety will be done. Bureau of Indian Standards (BIS), Economic Commission for Europe (ECE), and the U.S. Department of Transportation (DOT) are the most common cases. They will ensure that our helmets pass the standards provided for performance, safety, and longevity.

We will also connect our alarm system with law enforcement, ambulance dispatching centers, and the operational emergency services under the umbrella of 108 to guarantee rapid and effective answer actions in case of incidents.

The Environmental Influence of Sustainability

In order to achieve a lower carbon footprint, we will engage in green resource and green manufacturing processes.

Our helmet, wherever possible, shall be manufactured from recyclable or biodegradable material so as to avoid plastic usage.

Energy-efficient electronic devices will also be used to improve battery life and consequently minimize electrical waste.

We are also considering using power sources that capture kinetic energy or solar charging to reduce the utilization of non-renewable power even more. From these ethical, legal, and environmental factors, we will be designing a smart helmet that not only will be sustainable, responsible, and environmentally friendly, but will also provide riders' safety and their personal information as well as the environment.

Conclusion & Future Scope

The AI-Powered Smart Helmet, a revolutionary solution, is designed to enhance safety, help prevent injuries, and give timely emergency intervention. The helmet ensures proactive monitoring of the rider's health, environment, and riding behaviors by integrating AI, IoT, and smart sensors. The helmet is an important resource for adventure riders, workers in high-risk job sites, and motorcycle riding. Some features of this revolutionary technology include GPS tracking and connectivity, drowsiness detection and notification, and crash detection when the rider does not react. This technology is intended to not only reduce the risk of collision but guarantee a response by medical professionals during an emergency, ultimately benefiting all riders in terms of safety and saving a considerable number of lives.

Development of more advanced Artificial Intelligence models for prevention of accidents and real-time risk prediction. Behavior analysis through machine learning for improved rider safety. Integration with Smart Vehicles Engagement with self-driving vehicles to avoid a collision. More effective vehicle-to-Everything (V2X) connectivity to enhance road safety. Cloud Connectivity with 5G Faster information transfer for cloud-based accident analysis and safety monitoring. Software updates and remote diagnostics. Next-generation health monitoring AI-based internal and external monitoring using vital sign analysis and identification of potential issues before they become actual problems. Automated alerts to the nearest hospitals for emergency medical assistance. Integration of augmented reality (AR) AR-infused heads-up displays (HUD) for warnings of hazards and navigation. Obstacles and Limitations Potential Hazards: technological problems caused by obsolete or incompatible technology.

Financial hazards from inadequate investment and/or going over-budget. Challenges: legal and regulatory compliance issues (permits, data insurance protections, etc.). availability of resources (shortage of products/equipment and qualified personnel). Challenges for implementation: stakeholder resistance or lack of support. Cross-functional teams and coordination issues. Development TimeLine Research Stage - two to four months to obtain information, analyze feasibility, and finalize requirements. Prototyping stage: three to two phases of testing: two to three months for user feedback loops and quality control.

Phase of Implementation: 6–12 months for extensive deployment and milestone tracking